<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>6</td><td>(a)</td><td>(i)</td><td>$kx = mg$ (1)
$x = \frac{0.150 \times 9.81}{7.5}$ = 0.196 m answer (1)</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td>2</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.150}{7.5}}$ substitution (1)
= 0.889 s (1)
NB. No credit for use of $T = 2\pi\sqrt{\frac{l}{g}}$</td><td>1</td><td>1</td><td></td><td>2</td><td>2</td><td>2</td></tr>
<tr><td></td><td>(b)</td><td></td><td>Drop in amplitude at beginning and end compared, e.g. 0.033 m and 0.005 m over first and last intervals. [Or rates of decrease: 0.0033 m per oscillation and 0.0005 m per oscillation] consider rate of decrease (1)]
Justify by noting the magnitude is larger at the start than at end of experiment. (1)</td><td></td><td></td><td>2</td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>When n = 0, $A = A_0 e^{-\frac{0}{N}} = A_0 e^0 = A_0$
[Alternative: say $e^0 = 1$, so $A = A_0$.]</td><td></td><td></td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>$A = A_0 e^{-\frac{n}{N}}$; $0.029 = 0.095 e^{-\frac{30}{N}}$ substitution (1)
$N = \frac{-30}{\ln\left(\frac{0.029}{0.095}\right)}$ or equiv → N =25.28 ≅ 25 answer (1)</td><td>1</td><td>1</td><td></td><td>2</td><td>2</td><td>2</td></tr>
<tr><td></td><td>(d)</td><td>(i)</td><td><table><tr><th>Oscillation number (n)</th><th>Amplitude (A) /m</th><th></th></tr><tr><td>0</td><td>0.095</td><td>0</td></tr><tr><td>10</td><td>0.062</td><td>0.43</td></tr><tr><td>20</td><td>0.043</td><td>0.79</td></tr><tr><td>30</td><td>0.029</td><td>1.19</td></tr><tr><td>40</td><td>0.019</td><td>1.61</td></tr><tr><td>50</td><td>0.014</td><td>1.91</td></tr><tr><td>60</td><td>0.009</td><td>2.36</td></tr></table>All three values correct (1) [accept 1.6, 1.9, 2.4 if consistent]</td><td></td><td>1</td><td></td><td>1</td><td>1</td><td>1</td></tr>
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